

**강좌 키워드	Computer architecture, Hardware design, CPU implementation, RISC-V, Cache								
*1. 수업목표	This course introduces the basic principles of a modern computer system, and hardware and software design techniques and analysis related with computer architecture and organization, which is based on basic concepts of digital system design. This course will discuss how to design the control and datapath for a pipelined RISC processor, memory and storage, and multi/many-core systems. and emerging computer systems. The principles presented in lecture are reinforced in the laboratory through the practical exercises of system-level implementation (e.g., RISC-V based CPU implementation).								
**2. 교재 및 참고문헌	교재	Title: Computer Organization and Design RISC-V Edition (1st edition) Author: David Patterson John Hennessy							
	참고문헌								
**3. 강의계획	주요 수업방식	☐ 플립러닝 ☒ 이론 위주 수업 ☐ 토론 위주 수업 ☒ 프로젝트 수업 ☐ 기타							
	[Week 1] Introduction to Computer Architecture								
	[Week 2] Instruction Set Architectures (ISA)								
	[Week 3] ISA and RISC-V								
	[Week 4] Performance Evaluation								
	[Week 5] Single-Cycle CPU Design								
	[Week 6] Multi-Cycle CPU Design								
	[Week 7] Pipelined CPU Design								
	[Week 8] Midterm Exam								
	[Week 9] Pipelining: Data Hazards								
[Week 10] Pipelining: Control Hazards									
[Week 11] Pipelining: Branch Prediction and Exceptions									
[Week 12] Memory Hierarchy and Cache									
[Week 13] Virtual Memory									
[Week 14] I/O Systems & Introduction to Parallel Processing									
[Week 15] Final Exam									
● We will have four extra lab session (details will be announced later)									
*4. 평가방법	성적부여 방식	절대평가							
	등급제 여부	A~F							
	구분	출석	과제	중간	기말	수시평가	태도	기타	합계
	비율	5%	30%	30%	35%	0	0	0	100%
	비고		HW 구현	지필 논술형/ 주관식	지필 논술형/ 주관식				
	출석 규정	수업일수의 1/3을 초과하여 결석하면 성적은 "F" 또는 "U"가 됨(학칙 85조) 결석에 대하여 교원에게 별도로 출석인정을 받은 경우 예외로 할 수 있음 (학업성적처리규정. 조기취업자 출석 및 성적처리 지침)							
	기타 사항	부정행위 규정, 대체시험 여부 및 방법, 과제 또는 시험 피드백 제공 여부 등 평가방법 관련 기타 사항 - 수강신청 변경 기간 이후 수강취소 시 교원 승인으로 일괄 적용함. - 개강 2주 이후에는 수강취소 불가 (수강신청 변경 기간 중에는 동일하게 승인 없이 변경 가능)							
5. 정원 외 신청	추가 수용 인원								
6. 수강생 참고사항	선이수 교과목								
	수강 시 필요사항								
면담시간 및 장소		Any time based on an appointment							
7. 장애학생 지원사항 ※ 필요에 따라 내용 수정 가능	강의수강 관련	☐ 시각장애: 교재 제작(디지털교재, 점자교재, 확대교재 등), 대필도우미 허용 ☐ 지체장애: 교재 제작(디지털교재), 대필도우미 및 수업보조 도우미 허용 ☐ 청각장애: 대필 및 문자통역 도우미 활동 허용, 강의 녹취 허용 ☐ 건강장애: 질병 등으로 인한 결석에 대한 출석 인정, 대필도우미 허용 ☐ 학습장애: 대필도우미 허용 ☐ 지적장애/자폐성장애: 대필도우미 및 수업 멘토 허용							
	과제 및 평가 관련	☐ 시각장애/지체장애/청각장애/건강장애/학습장애: 과제 제출기한 연장, 과제 제출 및 응답 방식의 조정, 평가 시간 연장, 평가 문항 제시 및 응답 방식의							

		조정, 별도 고사실 제공 ○ 지적장애/자폐성장애: 개별화 과제 제출 및 대체 평가 실시
	비고	본 강의를 수강하는 장애학생들에게는 이상의 지원 서비스 이외에도 장애학생 개개인의 특성과 요구에 따라, 지도교수 및 장애학생지원센터와의 상담을 통하여 적절한 수준의 지원 서비스를 제공합니다. 장애학생에 대한 지원서비스와 관련하여 문의사항이 있는 학생들은 담당교수 *** (02-880-****) 혹은 장애학생지원센터(02-880-8787)로 문의바랍니다.

◎ 강의계획서 직접입력 시 필수 입력 항목: *, **

◎ 강의계획서 첨부파일 업로드 시 시스템 필수 입력 항목: **

**Course Keywords	Computer architecture, Hardware design, CPU implementation, RISC-V, Cache								
*1. Goals	This course introduces the basic principles of a modern computer system, and hardware and software design techniques and analysis related with computer architecture and organization, which is based on basic concepts of digital system design. This course will discuss how to design the control and datapath for a pipelined RISC processor, memory and storage, and multi/many-core systems. and emerging computer systems. The principles presented in lecture are reinforced in the laboratory through the practical exercises of system-level implementation (e.g., RISC-V based CPU implementation).								
**2. Reading Materials	Textbooks	Title: Computer Organization and Design RISC-V Edition (1st edition) Author: David Patterson John Hennessy							
	References								
**3. Course Schedule	Lecture Method	☐ Flipped learning ☒ Theory-driven ☐ Discussion-oriented ☒ Project-based ☐ Others							
	[Week 1] Introduction to Computer Architecture [Week 2] Instruction Set Architectures (ISA) [Week 3] ISA and RISC-V [Week 4] Performance Evaluation [Week 5] Single-Cycle CPU Design [Week 6] Multi-Cycle CPU Design [Week 7] Pipelined CPU Design [Week 8] Midterm Exam [Week 9] Pipelining: Data Hazards [Week 10] Pipelining: Control Hazards [Week 11] Pipelining: Branch Prediction and Exceptions [Week 12] Memory Hierarchy and Cache [Week 13] Virtual Memory [Week 14] I/O Systems & Introduction to Parallel Processing [Week 15] Final Exam ● We will have four extra lab session (details will be announced later)								
*4. Evaluation	Grading Method	Relative evaluation							
	Grading Type	A~F							
	Item	Attendance	Assignment	Midterm	Final	Quizzes	Attitude	Other	Total
	Rate	5%	30%	30%	35%	0	0	0	100%
	Note		HW implementation	Essay test	Essay test				
	Attendance Policy	Students who are absent more than 1/3 of class days will receive "F" or "U" grade. Students whose attendance is acknowledged can be exceptions. (Academic Grading Regulations, Guidance of Attendance and Grading for Early Employed Students)							
	Other	Other matters pertaining to the evaluation method such as regulations on cheating, whether and how alternative tests are made, and whether feedback for assignments or tests is provided - Any course withdrawal after the Add/Drop period will be processed only with the instructor's approval. - Withdrawal is not permitted after the second week of the semester. (During the Add/Drop period, changes can be made without approval.							
5. Quota Exceeding Course Registration	Capacity	Up to 00 Students							
6. Guideline for Students	Prerequisite Courses								
	Requirements								
	Office Hours		Any time based on appointment						
7. Support Services for Students	For Lectures	☐ Visual Impairment: Make textbooks(digital textbook, braille textbook, enlarged textbook etc.), Allow note takers ☐ Physical Disability: Make textbooks(digital textbook), Allow note takers and							

with Disabilities * Contents can be modified as needed		assistants ☐ Hearing Impairment: Allow note takers and translators, Allow lecture recording ☐ Health Impairment: Excuse absence due to health problems, Allow note takers ☐ Learning Disability: Allow note takers ☐ Intellectual Disability / Autism Spectrum Disorder: Allow note takers and mentors
	For Assignments & Evaluations	☐ Visual Impairment / Physical Disability / Hearing Impairment / Health Impairment / Learning Disability: Extend assignment deadlines, Offer alternate assignment submission and response method, Extend testing period, Offer alternate testing method, Offer different testing room ☐ Intellectual Disability / Autism Spectrum Disorder: Offer individualized assignments and alternative evaluations
	Other	Students who take this course can get appropriate level of support service including the support listed above depending on the students' individual characteristics and needs through consultation with professors and the Support Center for Students with Disabilities. If you have any questions concerning support service for students with disabilities you can contact Professor *** (02-880-****) or Support Center for Students with Disabilities (02-880-8787).

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